

Institute of Leather Engineering and Technology
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TANNERY WASTEWATER TREATMENT
Study Material for Undergraduate Students

CHAPTER 3

BIOLOGICAL TREATMENT

Removing the Dissolved Organic Load with Micro-organisms

Activated Sludge · Aeration · Trickling Filters · RBC · Biofilters

Anaerobic Reactors · Ponds and Lagoons · Nitrogen Removal

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3.1 Introduction: The Load That Only Life Can Remove

Chapter 1 removed what could be screened, floated and settled. Chapter 2 removed the colour, the refractory organics, the residual metals, the salts and the pathogens. Between them these two chapters deployed a formidable range of physical and chemical technology.

Yet one pollutant has survived both, and it is the largest single item remaining: the DISSOLVED BIODEGRADABLE ORGANIC MATTER, which after primary treatment still amounts to 800–1,200 mg/L of BOD against a discharge limit of 50 mg/L.

Why did the earlier chapters not remove it? Consider each option in turn:

- **It cannot be screened or settled** — it is dissolved, so there are no particles to intercept or to sink.
- **It could be adsorbed on carbon** — but at 1,000 mg/L the carbon would exhaust within hours, and the cost would be absurd. Carbon is for the small refractory fraction, not the bulk load.
- **It could be oxidised chemically** — but ozone at 30–100 mg/L per unit of COD would cost more than the leather is worth.
- **It could be rejected by a membrane** — but the membrane would foul organically within days.

There remains one method which removes this load cheaply, reliably and at enormous scale, and it is not an invention of engineers at all. In every river on earth, bacteria consume organic matter as food. Biological treatment is that natural process brought indoors, concentrated and controlled.

KEY IDEA

Physical processes remove what can be separated. Chemical processes remove what can be transformed or transferred. Biological processes remove what can be EATEN — and for the bulk dissolved organic load of a tannery effluent, eating is by far the cheapest method known.

3.1.1 What biological treatment achieves

Parameter	Entering the biological stage	Leaving it
BOD₅	800 – 1,200 mg/L	50 – 150 mg/L
COD	2,500 – 4,000 mg/L	200 – 500 mg/L
Suspended solids	400 – 800 mg/L	50 – 150 mg/L
Ammonia nitrogen	High	Low, if the plant nitrifies

Parameter	Entering the biological stage	Leaving it
Dissolved solids	Unchanged	Unchanged

Table 3.1 Typical performance of biological treatment on tannery effluent.

A well-designed biological stage removes 85–95% of the BOD. The last row is repeated from earlier chapters only as a reminder: biological treatment does not remove salt, and never will.

3.2 The Biological Principle

3.2.1 The idea in plain words

Micro-organisms need food and they need energy. The dissolved organic matter in wastewater is, from their point of view, an abundant supply of both. Given the right conditions they consume it, multiply, and in doing so remove the pollution from the water.

The engineering consists of three steps, and the third is the one students most often forget.

1. **Assemble a dense population of micro-organisms** — far denser than anything found in a river.
2. **Feed them the wastewater and supply what they need** — oxygen, in the aerobic case — so that they consume the organic matter rapidly.
3. **Separate the micro-organisms from the water and remove them.** This is essential, because the pollution has not been destroyed: it has been CONVERTED into microbial cells. If those cells leave with the effluent, the pollution has merely changed from a dissolved form into a suspended one.

KEY IDEA

Biological treatment is microbial farming. You grow micro-organisms, feed them the pollution, and then harvest them as sludge. The pollution leaves the plant inside the microbes — which is why every biological process needs a solids-separation step, and why every biological process produces sludge.

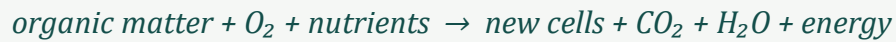
3.2.2 A formal definition

DEFINITION

Biological treatment is the use of micro-organisms, under controlled conditions, to convert the dissolved and colloidal biodegradable organic matter of wastewater into microbial biomass and stable end products — carbon dioxide and water aerobically, or biogas anaerobically — followed by the separation of that biomass from the treated water.

3.2.3 The overall reaction

In aerobic conditions the whole of biological treatment can be summarised in one word equation:



with bacteria acting as the catalyst throughout

Two products are harmless and leave by themselves: carbon dioxide escapes to the atmosphere, and water stays in the effluent. The third — new cells — is the sludge that must be removed. This single equation explains why every activated sludge plant in the world produces sludge, and why sludge handling is an unavoidable cost of biological treatment.

The equation also tells us what must be supplied. Oxygen must be provided continuously, which is why aeration (Section 3.8) is the largest energy cost of a treatment plant. And nutrients — principally nitrogen and phosphorus — must be present, because cells cannot be built from carbon alone.

THE NUTRIENT RULE

For healthy aerobic growth the wastewater should contain approximately BOD : N : P = 100 : 5 : 1 by mass. Anaerobically, far less nitrogen is needed — roughly COD : N : P = 350 : 5 : 1 — because so little biomass is produced.

Tannery effluent is usually rich in nitrogen (from deliming and from hide protein) but may be short of phosphorus. A nutrient deficiency causes weak, poorly-settling floc, and it is an easily-missed cause of plant failure.

3.2.4 What biological treatment requires of the earlier stages

Because the treatment agent is ALIVE, it can be injured — and this makes biological treatment uniquely dependent on what happens upstream.

Requirement	Why, and where it is provided
A biodegradable load	Micro-organisms consume only what they can digest. The refractory fraction is Chapter 2's business
A non-toxic influent	Sulphide and chromium poison the biomass. Both are removed in Chapter 1 — sulphide by oxidation, chromium by recovery
A steady flow and load	Micro-organisms adapt slowly; a sudden slug can destroy in an hour a biomass that took weeks to build. This is what equalisation (Chapter 1) exists for

Requirement	Why, and where it is provided
A tolerable pH	Most organisms work best at pH 6.5–8.5; neutralisation in Chapter 1
Adequate nutrients	The 100 : 5 : 1 ratio above
Tolerable salinity	The tannery's chronic difficulty — see Section 3.18

This is worth stating plainly: MOST FAILURES OF BIOLOGICAL TREATMENT ARE NOT FAILURES OF THE BIOLOGICAL STAGE. They are caused by something upstream that should have been removed, buffered or corrected before the wastewater ever reached the micro-organisms.

3.3 The Micro-organisms

The biological community in a treatment plant is a small ecosystem, and it is worth knowing who lives in it and what each does.

Organism	Role	Significance to the operator
Bacteria	The workhorses. They perform almost all the BOD removal, oxidising organic matter for the energy they need to grow	Their ability to form FLOCS is what makes settling possible
Protozoa	Single-celled grazers that eat free-swimming bacteria	They polish the effluent and make it clear. Active stalked ciliates under the microscope are the best single sign of a healthy sludge
Rotifers and higher organisms	Multicellular grazers appearing in older sludge	Indicate a long sludge age and a stable, well-oxidised system
Fungi	Usually minor	Become dominant at low pH or when nitrogen is deficient; their filamentous growth causes sludge that will not settle
Algae	Photosynthetic; not present in tanks	Central in ponds (Section 3.13), where they supply the oxygen
Methanogenic archaea	Produce methane from acetate and hydrogen	The slow, sensitive organisms at the end of the anaerobic chain (Section 3.14)

Table 3.2 The organisms of biological treatment.

3.3.1 The floc: why it matters

An individual bacterium is about one micrometre across and would never settle in a clarifier. Treatment is only possible because bacteria aggregate into FLOCS — particles typically 50–200 μm across, held together by the sticky extracellular polymeric substances (EPS) that the bacteria themselves secrete, often with filamentous organisms forming an internal skeleton.

A good floc is dense and compact and settles quickly. A poor floc is light and open, traps water, and settles slowly or not at all. Since the biomass must be separated before the effluent can be discharged, the SETTLING quality of the floc is as important as the biological activity of the organisms.

KEY IDEA

A biological plant has two duties of equal importance: to REMOVE the organic matter, and to SEPARATE the biomass afterwards. Excellent biology with poor settling produces a failing effluent.

3.3.2 A practical consequence

Because the community changes with operating conditions, the microscope is a diagnostic instrument. An operator who examines the sludge regularly learns to read its condition — the presence of stalked ciliates, the appearance of filaments, the death of protozoa — long before the laboratory results arrive. Section 3.9 develops this.

3.4 Classification of Biological Processes

A student meeting this subject is confronted with a bewildering list of names: activated sludge, extended aeration, oxidation ditch, trickling filter, rotating biological contactor, aerated lagoon, waste stabilisation pond, moving bed biofilm reactor, upflow anaerobic sludge blanket. It appears to be a catalogue to be memorised.

It is not. Every one of them rests on the biology of Section 3.2, and they differ in only TWO respects. Learn the two questions and the catalogue becomes a map.

3.4.1 First question: is oxygen present?

Micro-organisms need something to accept the electrons released as they break down organic matter. If dissolved oxygen is available they use it, and the process is AEROBIC. If it is not, other organisms use other acceptors.

Feature	Aerobic	Anaerobic
Electron acceptor	Dissolved oxygen	Carbon dioxide ($\rightarrow$ methane)

Feature	Aerobic	Anaerobic
End products	CO ₂ + H ₂ O + much new biomass	Biogas (CH ₄ + CO ₂) + little biomass
Energy	Consumes energy (aeration)	Produces energy (methane)
Sludge yield	0.4 – 0.6 kg VSS/kg BOD	0.05 – 0.15 kg VSS/kg COD
Speed	Fast (hours)	Slow (days)
Best suited to	Medium-strength wastewater	High-strength wastewater
Effluent quality	Can meet discharge limits	Partial — needs aerobic polishing

Table 3.3 The first division: aerobic and anaerobic treatment.

The difference in sludge yield deserves an explanation, because students often ask why anaerobic processes produce so little sludge. The answer is energetic. Aerobic oxidation releases a great deal of energy per unit of substrate, and abundant energy permits abundant synthesis of new cells. Anaerobic conversion to methane releases far less, because most of the energy remains locked in the methane. With less energy available, fewer cells can be built.

A third condition must be named now, because it becomes important in Section 3.15:

Anoxic. dissolved oxygen absent but NITRATE present, so that bacteria use nitrate as the electron acceptor. Students frequently confuse anoxic with anaerobic. The distinction is simple: anoxic has nitrate; anaerobic has neither oxygen nor nitrate.

3.4.2 Second question: where do the micro-organisms live?

The biomass must be kept in the reactor while the water flows through. There are two ways to arrange this.

Suspended growth. the micro-organisms float freely in the wastewater as flocs, forming a mixture called MIXED LIQUOR. Water and biomass travel together, so the biomass must afterwards be separated by settling and returned to keep the population up. Activated sludge and its variants work this way.

Attached growth. the micro-organisms grow as a BIOFILM — a slime layer fixed to a solid surface such as stone, plastic media or rotating discs. The wastewater flows past and the pollutants diffuse into the film. Because the biomass is physically fixed it cannot wash out, so no sludge return is needed and the settling problems largely disappear.

Hybrid systems. combine the two — plastic carriers are added to a suspended-growth tank so that flocs and biofilm work side by side.

Feature	Suspended growth	Attached growth
Biomass retention	By clarifier and sludge return	Inherent — cannot wash out
Sludge age control	Set by the operator (wasting rate)	Self-regulating by sloughing
Risk of bulking	Yes — the main failure mode	Essentially none
Tolerance of shock and toxicity	Lower — biomass fully exposed	Higher — the film protects itself
Operator skill required	High	Low to moderate
Flexibility of control	High — several parameters	Limited
Footprint	Compact	Larger for classical units

Table 3.4 The second division: suspended and attached growth.

3.4.3 The classification grid

	Suspended growth	Attached growth
Aerobic	Activated sludge and all variants; oxidation ditch; SBR; aerated lagoon; membrane bioreactor	Trickling filter; rotating biological contactor; MBBR; biofilter
Anaerobic	Anaerobic digester; UASB reactor; anaerobic pond	Anaerobic filter; anaerobic baffled reactor

Table 3.5 The classification grid. Every named process belongs in one of these four boxes.

KEY IDEA

Every biological process is a combination of two choices: aerobic or anaerobic, and suspended or attached growth. There are only four possibilities, and every process name in this chapter is a variation within one of them.

3.4.4 A third distinction: mechanised and natural

Cutting across the grid is a practical distinction that matters greatly in Bangladesh. MECHANISED systems (activated sludge, trickling filters, MBBR) use machinery to intensify the process into a small volume. NATURAL systems (ponds, lagoons, aquatic plant systems) use land, sunlight and time instead.

Neither is superior in the abstract. The choice reduces to four questions that recur throughout this chapter: what effluent quality is required; how much land is available; how much energy and money can be spent; and how skilled is the workforce that will run the plant every day? A sophisticated plant that nobody can operate is worse than a simple one that works.

3.5 Growth, Substrate Use and the Control Parameters

Before studying individual processes we need the language in which they are designed and controlled. This section introduces it once; every later section uses it.

3.5.1 The growth curve

If micro-organisms are placed in a vessel with a fixed quantity of food, their mass follows a curve with four recognisable phases.

Phase	What happens	Significance
Lag	Organisms adapt and make enzymes; no measurable growth	The delay when starting a plant, or recovering from a toxic shock
Exponential (log)	Growth at the maximum rate; food abundant	High-rate processes operate here
Stationary	Growth balanced by death; food becoming limiting	Conventional plants operate near here
Endogenous (decay)	Food exhausted; cells consume their own material and the mass falls	Extended aeration exploits this deliberately

Table 3.6 The four phases of microbial growth.

The whole art of operating a biological process consists in choosing which phase the organisms shall be held in, and holding them there. That choice is made through the parameters below.

3.5.2 Measuring the biomass

MLSS — Mixed Liquor Suspended Solids. all the suspended solids in the aeration tank: living cells, dead cells and inert matter together. Found by filtering a sample and drying at 105 °C. Typically 1,500–4,000 mg/L.

MLVSS — Mixed Liquor Volatile Suspended Solids. the organic fraction of the MLSS, found by igniting the dried residue at 550 °C and measuring the weight lost. Since living matter is organic, MLVSS represents the ACTIVE BIOMASS. Typically 70–80% of MLSS.

$$MLVSS \approx 0.70 \text{ to } 0.80 \times MLSS$$

‘Volatile’ here means ‘burns off at 550 °C’ — not easily evaporated

Both are needed. MLVSS is the true measure of biomass, but it requires a furnace and several hours; MLSS needs only a filter and an oven. Plants therefore measure MLSS daily and convert using the ratio established for their own sludge.

3.5.3 The four control parameters

These four quantities, with dissolved oxygen, are what an operator actually adjusts. They are introduced here in the order in which each becomes possible to define.

(a) HRT — hydraulic retention time

How long the WATER stays in the reactor:

$$HRT = V / Q$$

V = reactor volume (m³); Q = flow (m³/d). Typically 4–8 h for conventional activated sludge

Rearranged as $V = Q \times HRT$, this is also the equation that sizes the tank.

(b) SRT — solids retention time, or sludge age

How long the MICRO-ORGANISMS stay in the system — the mass of biomass held, divided by the mass removed each day:

$$SRT = (V \times X) / (Q_{vv} \times X_{vv})$$

X = MLSS in the reactor; Q_{vv} and X_{vv} = flow and concentration of waste sludge. Typically 5–15 d

Now compare (a) and (b), because their relationship is the central insight of biological treatment. The water stays 4–8 HOURS; the micro-organisms stay 5–15 DAYS. They are decoupled, and the reason is the return sludge line: the clarifier catches the biomass and returns it while the water flows on.

KEY IDEA

SRT $\gg$ HRT. The sludge is retained for DAYS while the water passes through in HOURS. This decoupling is what allows a small tank to hold a large, mature microbial population.

It has a further consequence of the first importance: SRT SELECTS THE SPECIES. Any organism that reproduces more slowly than the sludge age is washed out faster than it can multiply. This is why nitrification (Section 3.15) requires a sludge age of about ten days — the nitrifying bacteria simply cannot reproduce fast enough at shorter sludge ages.

(c) F/M — the food-to-microorganism ratio

How much food each unit of biomass receives per day:

$$F/M = (Q \times S_0) / (V \times X_v)$$

S_0 = influent BOD (kg/m^3); X_v = MLVSS (kg/m^3). Units: kg BOD per kg MLVSS per day

Every symbol here has now been defined, which is why F/M is introduced third. F/M and SRT move in opposite directions: a large biomass consuming little food per head (low F/M) is also a biomass wasted slowly and therefore kept a long time (high SRT). They are two ends of one dial.

Setting	Sludge character	Result
High F/M 0.4–1.5 Short SRT 2–5 d	Young, fast-growing, dispersed	More sludge; poorer settling; no nitrification; higher effluent BOD
Normal F/M 0.2–0.5 SRT 5–15 d	Well-formed flocs	Reliable BOD removal; predictable sludge
Low F/M 0.05–0.15 Long SRT 20–30 d	Old, mineralised, endogenous	Very little sludge; stable; nitrifies; larger tank

Table 3.7 The F/M – SRT dial and its consequences.

(d) SVI — sludge volume index

Not a control input but a DIAGNOSTIC: it tells the operator whether the biomass produced by the settings above will actually separate. It is found from the SV_{30} test — one litre of mixed liquor settled for thirty minutes in a graduated cylinder:

$$SVI = SV_{30} (\text{mL}/\text{L}) \times 1000 / MLSS (\text{mg}/\text{L})$$

SVI is the volume in mL occupied by one gram of dry sludge after 30 minutes of settling

SVI (mL/g)	Interpretation	Consequence
< 100	Good, dense floc	Clarifier works well; thick return sludge
100 – 150	Fair	Acceptable, but watch the trend
> 150	Bulking sludge	Blanket rises; solids carry over; effluent fails

3.5.4 Summary of the parameters

Parameter	Formula	Typical value	Controlled by
MLSS	measured (105 °C)	1,500 – 4,000 mg/L	Wasting rate
MLVSS	measured (550 °C)	$0.7 - 0.8 \times \text{MLSS}$	—
HRT	V / Q	4 – 8 h	Tank volume
SRT	$V \cdot X / (Q_{vv} \cdot X_{vv})$	5 – 15 d (≥ 10 to nitrify)	Wasting rate
F/M	$Q \cdot S_0 / (V \cdot X_v)$	0.2 – 0.5	MLSS and volume
Dissolved oxygen	measured	≈ 2 mg/L	Blower output
SVI	$SV_{30} \times 1000 / \text{MLSS}$	< 100 mL/g	Diagnosed, not set
Return sludge ratio	Q_r / Q	50 – 100%	RAS pump

Table 3.8 The control parameters of biological treatment.

KEY IDEA

The sludge WASTING RATE controls both the MLSS and the SRT, and through them the F/M ratio. It is the single most powerful lever the operator possesses. Waste more sludge and MLSS falls, SRT shortens and F/M rises; waste less and the opposite occurs.

3.6 The Activated Sludge Process

3.6.1 The problem it solved

The obstacle to treating wastewater in a tank of reasonable size is arithmetic. If wastewater passes through in six hours, and the bacteria inside need several days to build a useful population, the bacteria are washed out faster than they can grow. The tank never accumulates enough biomass to do the work.

The solution was found in 1914 at Manchester by Arden and Lockett. They aerated sewage in a batch, let the solids settle, poured off the treated water, and then — crucially — KEPT THE SETTLED SLUDGE and added the next batch of sewage to it. With each repetition treatment became faster, because the retained sludge contained a dense, adapted population of hungry micro-organisms. It had become biologically ACTIVATED, and the process took its name from that.

KEY IDEA

The whole invention lies in one idea: RETAIN THE BIOMASS AND RETURN IT. The water passes through in hours; the micro-organisms stay for days. Every later refinement in this chapter is a variation on that principle.

3.6.2 Biological principle and microorganisms

The organisms are the mixed bacterial community of Section 3.3, together with the protozoa and rotifers that graze on them. They are held in suspension as flocs in the mixed liquor. The removal of BOD proceeds in three overlapping stages, and it is useful to separate them.

1. **Adsorption.** Within minutes of contact, colloidal and particulate organic matter is adsorbed onto the sticky floc surface. This causes a rapid initial fall in BOD which is physical rather than metabolic — the organic matter is captured but not yet destroyed.
2. **Oxidation and synthesis.** The micro-organisms then metabolise what they have captured. Part is oxidised to carbon dioxide and water to release energy; part is converted into new cell material. This is where true removal occurs, over several hours.
3. **Endogenous respiration.** When the food supply runs low, the organisms consume their own cell substance for maintenance energy. This reduces the mass of sludge produced and stabilises what remains.

The distinction between the first two stages explains why contact time matters. Too short a contact and only adsorption has occurred: the BOD appears to have fallen, but the organic matter is still there, held on the floc, and will exert its demand later.

3.6.3 Process configuration

A conventional activated sludge plant has four elements.

Component	Function
Aeration tank (reactor)	Settled wastewater is mixed with a dense population of micro-organisms and aerated continuously. Air both supplies oxygen and keeps the solids in suspension. Within 4–8 hours most of the dissolved organic matter has been absorbed and oxidised
Secondary clarifier	The mixed liquor flows to a quiescent tank where flocs settle and clarified water overflows. It performs two duties at once: it CLARIFIES the effluent and THICKENS the sludge
Return sludge line (RAS)	Settled sludge is pumped continuously from the clarifier floor back to the aeration tank inlet. This is the step that retains the biomass. Typical rates are 50–100% of the incoming flow

Component	Function
Waste sludge line (WAS)	Because the organisms multiply daily, a controlled quantity is removed each day to keep the population steady. This is the operator's principal control

Table 3.9 Components of the activated sludge process.

The flow of water and of sludge may be summarised:

Influent → Aeration tank → Secondary clarifier → Effluent

with sludge returning from the clarifier to the aeration tank (RAS), and a small stream wasted (WAS)

3.6.4 Design procedure

A design proceeds in a standard order, using the parameters of Section 3.5.

1. Establish the design flow Q and influent BOD S_0 , and hence the daily BOD load.
2. Fix the required effluent BOD and hence the removal efficiency.
3. Select the process modification, and from Table 3.10 choose design values of F/M (or SRT) and $MLSS$.
4. Calculate the reactor volume from the chosen F/M .
5. Check the resulting HRT and volumetric loading against normal ranges.
6. Calculate the sludge production, hence the wasting rate and the actual sludge age.
7. Calculate the oxygen requirement and size the aeration equipment.
8. Size the secondary clarifier and determine the return sludge rate.

Two working equations are needed. The reactor volume follows from rearranging the F/M definition:

$$V = (Q \times S_0) / (F/M \times X_v)$$

and the daily production of excess sludge is given by the observed yield:

$$P_x = Y_o^b \times Q \times (S_0 - S)$$

$Y_o^b \approx 0.4\text{--}0.6$ kg VSS per kg BOD removed at conventional sludge age, falling to 0.2–0.3 at extended aeration

WORKED EXAMPLE 3.1 — Design of an activated sludge plant

A tannery discharges 400 m³/d which, after primary treatment, has a BOD of 700 mg/L. The effluent BOD must not exceed 40 mg/L. Design a complete-mix activated sludge plant.

DESIGN DATA ADOPTED

$F/M = 0.30 \text{ kg BOD/kg MLVSS}\cdot\text{d}$; $MLSS = 3,000 \text{ mg/L}$; $MLVSS/MLSS = 0.75$

so $MLVSS = 2,250 \text{ mg/L} = 2.25 \text{ kg/m}^3$

Observed yield $Y_{o,b} = 0.45$; waste sludge concentration 8,000 mg/L

Step 1 BOD load applied = $0.700 \text{ kg/m}^3 \times 400 = 280 \text{ kg BOD/d}$

Step 2 BOD removed = $(700 - 40) = 660 \text{ mg/L} \rightarrow 0.660 \times 400 = 264 \text{ kg/d}$ (94.3%)

Step 3 Reactor volume $V = (Q \cdot S_0)/(F/M \times X_v) = 280 / (0.30 \times 2.25) = 415 \text{ m}^3$, say 420 m³

Step 4 HRT = $V/Q = 420/400 = 1.05 \text{ d} = 25 \text{ hours}$

COMMENT: far longer than the 4–8 h of a municipal plant, because the influent BOD is 700 mg/L instead of 200–300. The plant is in effect operating as extended aeration — which is entirely appropriate for tannery effluent.

Step 5 Volumetric loading = $280/420 = 0.67 \text{ kg BOD/m}^3\cdot\text{d}$ (0.3–1.0 range — acceptable)

Step 6 Sludge production $P_x = 0.45 \times 264 = 119 \text{ kg VSS/d}$

Step 7 Waste sludge flow = $119 / 8.0 \text{ kg/m}^3 = 14.9 \text{ m}^3/\text{d}$, say 15 m³/d

Step 8 Resulting sludge age:

biomass in tank = $420 \times 3.0 = 1,260 \text{ kg}$; wasted $\approx 120 \text{ kg/d}$

SRT = $1,260/120 = 10.5 \text{ days}$ — above 10 d, so nitrification can be expected

Step 9 Oxygen requirement:

carbonaceous $\approx 1.0 \times 264 = 264 \text{ kg O}_2/\text{d}$

if 30 mg/L ammonia N is oxidised: $0.030 \times 400 = 12 \text{ kg N/d} \rightarrow 4.57 \times 12 = 55 \text{ kg O}_2/\text{d}$

TOTAL = $319 \text{ kg O}_2/\text{d} = 13.3 \text{ kg/h}$; at $1.5 \text{ kg O}_2/\text{kWh} \rightarrow \text{blower} \approx 9 \text{ kW}$

Step 10 Secondary clarifier at $20 \text{ m}^3/\text{m}^2\cdot\text{d}$: area = $400/20 = 20 \text{ m}^2$.

Check solids loading with 100% return: $(400+400) \times 3.0 = 2,400 \text{ kg/d}$ over $20 \text{ m}^2 = 120 \text{ kg/m}^2\cdot\text{d}$ — at the upper limit.

Adopt $30 \text{ m}^2 \rightarrow$ diameter 6.2 m, giving $80 \text{ kg/m}^2\cdot\text{d}$. Adopt 6.5 m diameter.

COMMENT: the clarifier was governed by SOLIDS loading, not by overflow rate — as is usual in activated sludge.

Step 11 Return sludge ratio = $X/(X_r - X) = 3,000/(8,000 - 3,000) = 0.60$

$$Q_r = 0.60 \times 400 = 240 \text{ m}^3/\text{d}$$

SUMMARY: aeration tank 420 m^3 (HRT 25 h); MLSS $3,000 \text{ mg/L}$; F/M 0.30 ; SRT 10.5 d ; oxygen 319 kg/d , blower $\approx 9 \text{ kW}$; clarifier 6.5 m diameter; RAS $240 \text{ m}^3/\text{d}$; WAS $15 \text{ m}^3/\text{d}$.

3.6.5 Advantages and limitations

Advantages	Limitations
High and consistent BOD removal (85–95%)	High energy cost — continuous aeration
Compact compared with natural systems	Requires skilled daily operation
Highly flexible — many control parameters	Vulnerable to bulking and to toxic shock
Can be made to nitrify by extending sludge age	Produces a large volume of biological sludge
Well understood; abundant design experience	Performance depends entirely on settling

3.7 Modifications of the Activated Sludge Process

The basic process has been modified many times to suit different loads, land availability and effluent requirements. All share the design equations of Section 3.6; what differs is where the F/M – SRT dial is set, and how the tank is shaped.

Conventional (plug flow)

A long narrow aeration tank through which the wastewater passes in orderly sequence with little longitudinal mixing. Oxygen demand is highest at the inlet, where the food is strongest, and falls along the tank. Efficient, but the inlet region is vulnerable to shock loads.

Complete-mix

The tank is stirred so vigorously that its contents are uniform throughout. Any incoming slug of strong or toxic effluent is immediately diluted into the whole volume, which makes the process far more resistant to shock — the reason it is favoured for industrial wastewater.

Step feed and tapered aeration

Two solutions to the same problem of uneven oxygen demand along a plug-flow tank. STEP FEED introduces the influent at several points along the tank, spreading the demand. TAPERED AERATION leaves the feed at the head but installs more diffusers near the inlet and fewer towards the outlet, matching the air supply to the demand.

Extended aeration

The plant is run deliberately at very low F/M (0.05–0.15) and very long sludge age (20–30 d), so that the organisms are always short of food and endogenous respiration dominates. The result is a stable, forgiving process producing very little sludge, which nitrifies as a matter of course. The price is a much larger tank and higher aeration energy per unit of BOD removed.

Contact stabilisation

Exploits the rapid adsorption stage of Section 3.6.2. The wastewater is contacted with sludge for only 30–60 minutes — long enough for adsorption — after which the sludge is settled and oxidised in a separate, smaller stabilisation tank before being returned. Because only the sludge, not the whole flow, occupies the stabilisation tank, the total volume required is reduced.

Oxidation ditch

A continuous looped channel, oval in plan, around which the mixed liquor is circulated at 0.25–0.35 m/s by horizontal brush rotors that also aerate it. Kinetically it is an extended-aeration plant; hydraulically it behaves as a completely-mixed reactor because the contents circulate many times before leaving.

A feature of great practical value follows from the geometry. Just downstream of a rotor the dissolved oxygen is high; further round the loop it has been consumed and falls towards zero. The ditch therefore contains AEROBIC and ANOXIC zones in the same channel — which allows simultaneous nitrification and denitrification without any additional tankage (Section 3.15).

Sequencing batch reactor (SBR)

All operations occur in a single tank, separated in TIME rather than in space: FILL, REACT (aerate), SETTLE, DECANT, IDLE. No separate clarifier is needed, since the tank becomes its own settling basin — and a very good one, because settling occurs with no inflow to disturb it.

By switching the aeration on and off during the react phase, aerobic and anoxic conditions can be created in sequence, allowing nitrogen removal in one vessel. The penalty is a requirement for reliable automatic control, and usually two tanks alternating so that inflow can be continuous.

Modification	F/M	SRT (d)	HRT (h)	Distinguishing feature
Conventional plug flow	0.2 – 0.4	5 – 15	4 – 8	Load highest at inlet
Complete-mix	0.2 – 0.6	5 – 15	3 – 5	Dilutes shocks
Step feed	0.2 – 0.4	5 – 15	3 – 5	Feed split along tank
Extended aeration	0.05 – 0.15	20 – 30	18 – 36	Stable; least sludge; nitrifies
High-rate	0.4 – 1.5	2 – 5	1.5 – 3	Compact; partial treatment
Contact stabilisation	0.2 – 0.6	5 – 15	0.5 – 1*	Smaller total volume
Oxidation ditch	0.05 – 0.15	15 – 30	18 – 30	Looped channel; robust
SBR	0.05 – 0.3	10 – 30	cycle-based	One tank; good for nutrients

Table 3.10 Design ranges for activated sludge modifications. *contact time only.

Reading across this table repays study. Extended aeration and the oxidation ditch occupy the same region — low F/M, long sludge age, long retention — because they are essentially the same process in different shapes. The high-rate process is the opposite extreme, sacrificing effluent quality for compactness. The whole table is generated by moving two dials.

3.8 Aeration Systems

Purpose

To supply the oxygen that aerobic micro-organisms consume, and to keep the biomass in suspension.

3.8.1 The problem

Oxygen is very poorly soluble in water. At 20 °C, water in equilibrium with the atmosphere holds only about 9 mg/L — and a busy aeration tank can consume that entire amount in a few minutes. Oxygen must therefore be supplied continuously and cannot be stored in the water.

This makes aeration the defining mechanical operation of the activated sludge process and, typically, 45–60% of the total energy consumption of a treatment plant. Any improvement in aeration efficiency is money saved every day for the life of the works.

3.8.2 Two duties, not one

- **Oxygen transfer** — dissolving enough oxygen to satisfy the biological demand.
- **Mixing** — keeping the mixed liquor solids in suspension and in contact with the wastewater. If mixing fails, the biomass settles on the tank floor and goes anaerobic.

At low organic loading — in an extended-aeration plant, for instance — the mixing requirement may EXCEED the oxygen requirement, so the power input cannot be reduced below about 20–30 W/m³ even though the microbes need less air.

3.8.3 How oxygen transfer works

Oxygen moves from a gas bubble into the water across the gas–liquid interface, at a rate described by:

$$dC/dt = KLa (C_s - C)$$

KLa = overall transfer coefficient (1/h); C_s = saturation DO (mg/L); C = actual DO in the tank (mg/L)

Two lessons follow, and they explain much of practical operation.

- **The driving force is the DEFICIT (C_s – C).** The lower the dissolved oxygen in the tank, the faster oxygen dissolves. Running a tank at high DO is therefore doubly wasteful: it costs more air, and each unit of air transfers less oxygen. This is why plants are held at about 2 mg/L rather than at saturation.
- **KLa depends on the interfacial area.** For a given volume of air, many small bubbles present far more surface than a few large ones — the entire reason for fine-bubble diffusers.

3.8.4 Types of equipment

System	Description	Efficiency	Notes
Fine-bubble diffused	Porous or membrane diffusers on the floor; bubbles 2–5 mm	15 – 35% transfer	Most efficient; standard for new plants; needs cleaning
Coarse-bubble diffused	Simple orifices; bubbles > 6 mm	4 – 10%	Will not clog; used where mixing matters more
Surface aerator (vertical)	Impeller throws liquid into the air	1.2 – 2.4 kg O ₂ /kWh	Simple; no blower house; aerosols and heat loss

System	Description	Efficiency	Notes
Brush rotor (horizontal)	Rotating brush in a channel	1.2 – 2.0 kg O ₂ /kWh	Aerates AND propels the liquid; used in ditches
Jet / submerged turbine	Air injected beneath an impeller or into a jet	Moderate to high	Deep tanks; high demand; more maintenance

Table 3.11 Aeration systems compared.

3.8.5 How much oxygen is required

Two rules of thumb are accurate enough for preliminary work and easy to remember:

- **0.8 – 1.2 kg O₂** per kg of BOD removed — the carbonaceous demand.
- **4.57 kg O₂** per kg of ammonia nitrogen oxidised — the nitrogenous demand, if the plant nitrifies. Often rounded to 4.6.

The second figure surprises students: nitrifying a modest quantity of ammonia can add a quarter or more to the total air demand. It is one reason why nitrification is expensive to achieve.

WORKED EXAMPLE 3.2 — Oxygen and blower requirement

A tannery plant handles 500 m³/d. BOD entering the aeration tank is 600 mg/L and leaving is 40 mg/L. Ammonia nitrogen oxidised is 35 mg/L. Estimate the oxygen requirement and blower power.

Step 1 BOD removed = (600 – 40) = 560 mg/L = 0.560 kg/m³; × 500 = 280 kg BOD/d

Step 2 Carbonaceous demand at 1.0 kg O₂/kg BOD = 280 kg O₂/d

Step 3 Nitrogen oxidised = 0.035 × 500 = 17.5 kg N/d

Nitrogenous demand = 4.57 × 17.5 = 80 kg O₂/d

Step 4 TOTAL = 280 + 80 = 360 kg O₂/d = 15.0 kg O₂/h

Step 5 Air required: air is 23.2% oxygen by mass, density ≈ 1.20 kg/m³, so 1 m³ of air holds ≈ 0.278 kg O₂.

At 20% field transfer efficiency: 15.0 / (0.278 × 0.20) = 270 m³/h of air

Step 6 Blower power at 1.5 kg O₂/kWh: 15.0 / 1.5 = 10 kW

ANSWER: about 360 kg O₂/d, requiring roughly 270 m³/h of air and a blower of about 10 kW.

COMMENT: nitrification accounted for 80 of the 360 kg — more than a fifth — although the ammonia load was only 17.5 kg/d against 280 kg/d of BOD.

3.8.6 Controlling the aeration

Dissolved oxygen	Consequence
Below 0.5 mg/L	Treatment deteriorates; filamentous organisms thrive and cause bulking; odour develops
1 – 2 mg/L	Satisfactory for carbon removal
≈ 2 mg/L	The normal design set point; also sufficient for nitrification
Above 3 – 4 mg/L	No further benefit; energy wasted and transfer efficiency falls

Because oxygen demand varies through the day with the flow and strength of the wastewater, a plant whose blowers run at constant output wastes a great deal of energy at night. Simple automatic control from a dissolved-oxygen probe typically saves 20–30% of aeration energy and repays its cost quickly.

3.9 Operating Problems and Their Diagnosis

A design is only as good as the operation that follows. The characteristic problems of suspended-growth systems, and their remedies, are set out here.

3.9.1 Sludge bulking

The commonest and most damaging failure. Filamentous bacteria grow out from the flocs, producing a light, open structure that traps water and will not compact; the SVI rises above 150 mL/g. The sludge blanket rises in the clarifier, biomass carries over the weir, effluent suspended solids and BOD fail, and MLSS is progressively lost from the system.

The usual causes are low dissolved oxygen, prolonged low F/M, nutrient deficiency, septic influent containing sulphide, and wide swings in loading. The remedies should be applied in that order: correct the DO first, then the nutrients, then adjust F/M and SRT; consider a selector zone; and only as an emergency measure chlorinate the return sludge.

3.9.2 Diagnostic table

Symptom	Probable cause	Corrective action
SVI > 150; blanket rising	Filamentous bulking	Check DO, nutrients (100:5:1), F/M and SRT
Thick brown foam	Nocardia; grease; long SRT	Reduce SRT; remove foam physically; control oils
White billowing foam	Young sludge, low MLSS (start-up)	Reduce wasting and let MLSS build
Sludge rising in clarifier	Denitrification releasing N ₂ in the blanket	Increase sludge withdrawal; reduce detention
Pin-point floc; turbid effluent	F/M too low; sludge over-oxidised	Increase F/M by wasting more; shorten SRT
Dispersed growth; cloudy	F/M too high; overloading	Increase MLSS; reduce load
DO rises suddenly; sludge pale	TOXIC SHOCK — biomass killed	Find and stop the source; dilute; STOP wasting; reseed
Ammonia appears in effluent	Nitrification lost	Lengthen SRT beyond 10 d; check DO and alkalinity
Dark, septic, sulphide odour	Anaerobic conditions; low DO	Increase aeration; check blanket depth

Table 3.12 Diagnosis of activated sludge operating problems.

3.9.3 Toxic shock — the tannery problem

The greatest operational risk in a tannery plant deserves separate mention. A slug of chrome liquor, an un-oxidised sulphide-bearing lime dump, or a discharge of solvent or biocide can kill the biomass within hours.

The signs are distinctive and every operator should know them: the dissolved oxygen concentration RISES suddenly — because the microbes have stopped respiring and are no longer consuming it — the effluent BOD climbs, the sludge turns pale and disperses, and under the microscope the protozoa are dead or motionless.

The immediate response is to identify and stop the source, dilute the tank, maintain aeration, and above all STOP WASTING SLUDGE so as to preserve whatever biomass survives. Recovery requires reseedling with healthy sludge from another plant and gradual re-acclimatisation over days or weeks.

3.9.4 Routine monitoring

Frequency	Measurements
Every shift / daily	Dissolved oxygen; SV_{30} settling test; sludge colour and appearance; clarifier blanket depth; pH; temperature; flow
Daily to weekly	MLSS and MLVSS; influent and effluent BOD/COD; effluent suspended solids; calculate F/M, SRT and SVI; record RAS and WAS flows
Weekly to monthly	Microscopic examination of the sludge; nutrient check; ammonia and nitrate if nitrifying; sludge production trend

One habit is worth impressing above all others: record the TREND, not merely the number. A single SVI value means little, but an SVI that has climbed steadily for five days is an unmistakable warning of bulking, and gives the operator time to act before the effluent fails.

3.10 Trickling Filters

3.10.1 The idea: the biofilm

Anyone who has walked on the wet stones of a riverbed knows the slippery film that coats them. That film is a community of micro-organisms attached to the stone, feeding on organic matter carried past by the flowing water. A trickling filter is an engineered riverbed: a bed of media over which wastewater is distributed so that it flows past an enormous area of biological film.

Biofilm. a layer of micro-organisms attached to a solid surface and bound together in a matrix of extracellular polymeric substances (EPS) — the slime the organisms secrete. Typically 0.1–2 mm thick, containing bacteria, protozoa, rotifers, fungi, worms and insect larvae.

The name is misleading and should be corrected at once: a trickling filter does not filter. It is a biological reactor in which the medium provides SURFACE AREA, not straining.

3.10.2 How the biofilm works

Because the organisms cannot move to the food, the food must move to them, and it does so by DIFFUSION. This governs everything about biofilm behaviour.

1. **Transport into the film.** Organic matter and dissolved oxygen cross a thin stagnant liquid layer at the film surface and diffuse into the film. This transport step is frequently the rate-limiting one — the process is slowed not by the microbes' appetite but by how fast food can reach them.

2. **Layered metabolism.** Oxygen is consumed as it penetrates, and typically penetrates only 0.1–0.2 mm. A thick film therefore develops an aerobic outer zone doing most of the work, an anoxic zone beneath, and an anaerobic zone at the base.
3. **Sloughing.** As the film thickens, the anaerobic base receives no food, weakens and loses adhesion, and patches detach and wash away. This is the biofilm's equivalent of sludge wasting — except that it happens BY ITSELF, without the operator's intervention.

Two conclusions follow. A secondary clarifier is still needed, because the sloughed film (HUMUS SLUDGE) leaves with the effluent. And a thicker film does not mean better treatment: what matters is the surface area available and the contact between water and film.

KEY IDEA

In activated sludge the operator chooses the sludge age by deciding how much to waste. In a biofilm the system regulates its own age by sloughing. This is why biofilm plants need less skilled attention — and why they offer the operator less control.

3.10.3 Configuration and components

Component	Function
Media	A bed of rock or plastic providing the surface for biofilm. The heart of the filter
Rotating distributor	Two or four arms spraying settled wastewater evenly over the surface, usually driven by the reaction of the jets themselves. Even distribution is essential: dry patches lose their film and breed flies
Underdrain	A false floor collecting treated water and conveying it to the clarifier. Its second function is equally important — it ADMITS AIR, and must never run more than half full
Ventilation	Air moves by natural draught, driven by the temperature difference between air and water. No blower is needed in a classical filter
Secondary clarifier	Settles the sloughed humus sludge

The media

Property	Rock	Plastic
Specific surface area	45 – 55 m ² /m ³	90 – 200 m ² /m ³
Void space	40 – 50%	> 90%
Practical depth	1 – 2.5 m (weight-limited)	6 – 12 m (towers)

Property	Rock	Plastic
Clogging tendency	Higher	Low
Cost	Low where stone is local	Higher

Table 3.13 Filter media compared.

Plastic media transformed the trickling filter. Three to four times the surface area in the same volume, together with far greater void space, permitted tall towers carrying high organic loads with reliable ventilation — impossible with stone, whose weight limits depth and whose small voids clog.

3.10.4 Recirculation

Part of the filter effluent is commonly returned to the head of the filter. The RECIRCULATION RATIO $R = Q_r/Q$ usually lies between 0.5 and 3.0, and it performs four services:

- Dilutes a strong influent so the biofilm at the top is not overloaded.
- Maintains continuous wetting, preventing dry patches even at night when flow falls.
- Increases hydraulic flushing, sloughing excess film before it clogs the voids.
- Gives the wastewater a second and third pass over the biofilm.

Recirculation is thus the principal operating adjustment available — in effect the filter's only dial, in contrast to the several available on an activated sludge plant.

3.10.5 Design

Trickling filters are designed on two loading rates:

$$\text{Hydraulic loading} = (Q + Q_r) / A \quad (\text{m}^3/\text{m}^2 \cdot \text{d})$$

$$\text{Organic loading} = Q S_0 / V \quad (\text{kg BOD}/\text{m}^3 \cdot \text{d})$$

Class	Hydraulic ($\text{m}^3/\text{m}^2 \cdot \text{d}$)	Organic (kg BOD/ $\text{m}^3 \cdot \text{d}$)	BOD removal	Recirculation
Low-rate (standard)	1 – 4	0.08 – 0.32	80 – 90%	Usually none
Intermediate-rate	4 – 10	0.24 – 0.48	50 – 70%	Sometimes
High-rate	10 – 40	0.32 – 1.0	65 – 85%	Always
Super-rate (plastic)	12 – 70	0.8 – 6.0	65 – 85%	Usually
Roughing filter	40 – 200	1.5 – 6.0+	40 – 65%	Not required

Table 3.14 Classification of trickling filters by loading.

The last entry is directly relevant to tannery practice. A ROUGHING FILTER is loaded so heavily that it removes only 40–65% of the BOD — which sounds like failure until one realises that this is exactly its purpose. Placed ahead of a biological stage that could not otherwise cope, it knocks the peak off a very strong industrial effluent at low energy cost.

For estimating performance the NRC formula, developed by the United States National Research Council, remains the standard teaching tool for a single-stage rock filter:

$$E = 100 / [1 + 0.44 \sqrt{(W / (V \cdot F))}]$$

E = BOD removal (%); W = BOD load (kg/d); V = media volume (m³); F = recirculation factor

$$F = (1 + R) / (1 + 0.1 R)^2$$

WORKED EXAMPLE 3.3 — Trickling filter design

Design a rock-media trickling filter for 600 m³/d of settled wastewater of BOD 300 mg/L, to produce 60 mg/L. Adopt R = 1.5 and a depth of 2.0 m.

Step 1 BOD load $W = 0.300 \times 600 = 180$ kg/d

Step 2 Required efficiency $E = (300 - 60)/300 \times 100 = 80\%$

Step 3 Recirculation factor $F = (1 + 1.5)/(1 + 0.15)^2 = 2.5/1.3225 = 1.89$

Step 4 From the NRC formula:

$$80 = 100 / [1 + 0.44\sqrt{(180/(1.89 V))}]$$

$$1 + 0.44\sqrt{(95.2/V)} = 1.25$$

$$0.44\sqrt{(95.2/V)} = 0.25 \rightarrow \sqrt{(95.2/V)} = 0.568$$

$$95.2/V = 0.3227 \rightarrow V = 295 \text{ m}^3, \text{ say } 300 \text{ m}^3$$

Step 5 Plan area $A = V/\text{depth} = 300/2.0 = 150 \text{ m}^2$

$$\text{Diameter} = \sqrt{(4 \times 150/\pi)} = 13.8 \text{ m, adopt } 14 \text{ m}$$

Step 6 Check loadings:

$$\text{organic} = 180/300 = 0.60 \text{ kg BOD/m}^3 \cdot \text{d}$$

$$\text{hydraulic} = (600 + 900)/150 = 10.0 \text{ m}^3/\text{m}^2 \cdot \text{d}$$

Both place the unit in the HIGH-RATE class, consistent with the recirculation adopted. The design is self-consistent.

ANSWER: a filter 14 m in diameter and 2.0 m deep (300 m³ of media), recirculating 900 m³/d, followed by a clarifier for the humus sludge.

3.10.6 Operating problems

Problem	Cause	Remedy
Ponding	Excess biofilm or debris clogging the voids	Increase recirculation and flushing; rake or flood the surface; improve screening
Odour	Anaerobic conditions from overloading or blocked ventilation	Reduce organic loading; clear underdrains
Filter flies (Psychoda)	Dry patches where the film is thin	Continuous even wetting; higher recirculation; periodic flooding
Uneven treatment	Blocked distributor orifices; short-circuiting	Clean the arms; check the distributor level
Cold-weather loss	Low temperature reduces biology and draught	Cover the filter; reduce recirculation in winter

3.10.7 Advantages and limitations

Advantages	Limitations
Very low energy — natural draught, no blowers	Larger footprint than activated sludge
Simple to operate; little skilled attention	Little process control — recirculation only
Tolerates shock and toxic loads well	Ponding, odour and fly nuisance if neglected
Biomass cannot wash out; no bulking	Effluent quality slightly lower and less consistent
Produces less sludge than activated sludge	Performance falls in cold weather

3.11 Rotating Biological Contactors

Principle and configuration

The RBC applies the same biofilm principle in a different mechanical form. A series of large plastic discs, typically 2–3.7 m in diameter and closely spaced, is mounted on a horizontal shaft set in a contoured tank. About 40% of each disc is submerged, and the shaft rotates slowly at 1–2 rpm.

The elegance lies in what the rotation accomplishes. As each part of the disc dips into the wastewater, the biofilm on it absorbs organic matter; as the same part rises into the air, the film takes up oxygen. The rotation therefore alternately FEEDS and AERATES the biofilm, with no blower at all. The shear of rotation also strips off excess film, which passes to a downstream clarifier.

Staging

Units are arranged in three or four stages in series. The first stage receives the strongest load and does the bulk of the carbon removal; later stages, working on progressively weaker wastewater, polish the effluent and can nitrify. The first-stage loading is checked separately, because it is where overload occurs — most RBC failures begin with an overloaded, oxygen-starved lead stage.

Design

The RBC is designed on the surface area of the discs, since that is where the biomass lives:

$$A = Q S_0 / L_s$$

A = total disc area (m²); L_s = surface organic loading (g BOD/m²·d)

Duty	Loading (g BOD/m ² ·d)	Note
Carbon removal	4 – 10	Standard design
Carbon plus nitrification	2 – 5	Lower loading to admit the slow nitrifiers
First-stage maximum	≤ 30 (soluble BOD)	Exceeding this causes overload and odour

WORKED EXAMPLE 3.4 — RBC design

Design an RBC for 250 m³/d of wastewater at BOD 280 mg/L, with nitrification required. Use L_s = 4 g BOD/m²·d and standard shafts of 9,300 m².

Step 1 BOD load = $0.280 \times 250 = 70 \text{ kg/d} = 70,000 \text{ g/d}$

Step 2 Total disc area $A = 70,000/4 = 17,500 \text{ m}^2$

Step 3 Number of shafts = $17,500/9,300 = 1.88$, adopt 2 shafts ($18,600 \text{ m}^2$)

Step 4 Arrange as 4 stages: each shaft divided into 2 compartments by baffles

Step 5 Check the first stage: area = $17,500/4 = 4,375 \text{ m}^2$

loading = $70,000/4,375 = 16 \text{ g BOD/m}^2\cdot\text{d}$ — below the limit of 30. Acceptable.

ANSWER: 2 standard shafts baffled into 4 stages; first-stage loading $16 \text{ g/m}^2\cdot\text{d}$.

Advantages and limitations

Advantages	Limitations
Very low energy — only a slow drive motor	Shaft and bearing failures under heavy biofilm
Simple; little skilled attention required	Sensitive to overloading, especially the first stage
Compact compared with a trickling filter	Higher capital cost per unit capacity
Rapid start-up and recovery after upset	Disc replacement is expensive
Staging gives polishing and nitrification	Usually needs covering against weather and algae
No sludge return; no bulking	Performance falls in cold weather

3.12 Biofilters and Carrier Systems

3.12.1 The submerged biological aerated filter (BAF)

The word BIOFILTER is used in two senses, and students should distinguish them.

Submerged biological aerated filter (BAF). a reactor containing submerged granular or structured media on which biofilm grows, aerated from below, through which the wastewater passes. It performs biological treatment AND solids filtration in one unit, so no secondary clarifier is required. Media are typically expanded clay or plastic beads of 2–6 mm; loading rates are 2–8 kg BOD/ $\text{m}^3\cdot\text{d}$ — far higher than a trickling filter — and the unit is periodically backwashed like a sand filter.

Odour biofilter. a bed of organic packing through which foul air is passed, the odorous compounds being absorbed into the moist biofilm and oxidised. Used at pumping stations and covered tanks. This is air treatment, not water treatment.

The submerged BAF is the sense intended in a wastewater syllabus. Its attractions are a very small footprint, the elimination of the secondary clarifier, and good resistance to shock; its costs are the backwash system, the head loss and a more complex control regime.

3.12.2 Hybrid carrier systems: MBBR and IFAS

A hybrid process holds biomass in suspension AND on a fixed surface within the same reactor, to obtain the flexibility of suspended growth together with the robustness of attached growth.

MBBR — moving bed biofilm reactor

The tank is filled to 30–70% of its volume with small plastic carriers, kept in continuous motion by the aeration. Screens at the outlet retain them. Biofilm grows on the protected internal surface of the carriers, which offer 350–1,200 m²/m³.

There is no sludge return and no bulking, because the biomass stays on the carriers. Design is on carrier surface area — typically 5–15 g BOD/m²·d for carbon removal and 1–3 g/m²·d for nitrification. A simple solids separation still follows, to catch sloughed film.

IFAS — integrated fixed-film activated sludge

Carriers are added INTO an existing activated sludge tank, so that suspended flocs and attached biofilm work side by side, and the clarifier and return sludge line are retained.

The kinetic point is important: the suspended and attached fractions have DIFFERENT sludge ages. The suspended fraction is controlled by wasting as usual, but the attached fraction is retained indefinitely. This allows slow-growing organisms such as nitrifiers to persist regardless of how the suspended sludge is wasted — so an existing plant can be made to nitrify without extending its SRT.

WORKED EXAMPLE 3.5 — Uprating a plant with IFAS

An existing aeration tank of 600 m³ operates at MLVSS 2,500 mg/L treating 250 kg BOD/d at F/M 0.17. Production has grown and the load will rise to 400 kg BOD/d. Can carriers accommodate this without new tankage?

Step 1 Existing suspended biomass = $600 \times 2.5 = 1,500$ kg VSS

Step 2 Biomass needed at the new load, holding F/M at 0.17 = $400/0.17 = 2,353$ kg VSS

Additional biomass required = $2,353 - 1,500 = 853$ kg VSS

Step 3 Carriers of $500 \text{ m}^2/\text{m}^3$ carrying about $12 \text{ g VSS}/\text{m}^2$ hold $500 \times 12 = 6,000 \text{ g} = 6 \text{ kg VSS}$ per m^3 of carrier

$$\text{Carrier volume required} = 853/6 = 142 \text{ m}^3$$

Step 4 Fill fraction = $142/600 = 24\%$ — well within the practical limit of about 60%

Step 5 Aeration must be checked: oxygen demand rises from about 250 to about 400 $\text{kg O}_2/\text{d}$, and carriers need extra air for mixing

ANSWER: 142 m^3 of carriers (24% fill) accommodates a 60% increase in load within the existing tank. Retention screens and additional blower capacity are needed, but no new civil works.

COMMENT: this is why IFAS has become the standard answer for overloaded tannery plants and CETPs. Compare the cost of 142 m^3 of plastic carriers with that of building a second 400 m^3 aeration tank.

3.12.3 The membrane bioreactor (MBR)

An MBR replaces the secondary clarifier with a membrane (micro- or ultrafiltration) submerged in the aeration tank. The biology is that of activated sludge; what changes is the separation step.

The consequence is fundamental. Settleability ceases to be a design constraint — SVI, bulking and blanket control simply cease to matter — because separation is by filtration, not gravity. The biomass can therefore be held far more concentrated, at MLSS 8,000–12,000 mg/L instead of 2,000–4,000, so the same treatment occurs in roughly a third of the tank volume. The permeate is essentially free of suspended solids and bacteria, and is ready to feed reverse osmosis for reuse.

Against this stand membrane fouling requiring air scouring and periodic chemical cleaning, higher energy, membrane replacement every 5–10 years, and a requirement for skilled operation.

Note the classification point raised in Chapter 2: the MBR is treated here, among the biological processes, because its treatment FUNCTION is biological. The membrane is a separation device serving the biology, not a treatment process in its own right.

3.13 Ponds, Lagoons and Aquatic Plant Systems

3.13.1 The idea

Every process so far requires construction and machinery: concrete tanks, blowers, pumps, clarifiers, scrapers, and trained staff. For a large works this is justified. For a small tannery, or where capital and skilled operators are scarce, it may be prohibitive.

Ponds answer this by trading two expensive resources — energy and machinery — for two that are sometimes cheap: LAND and TIME. A pond is a basin excavated in the ground in which wastewater is retained for a long period while the same micro-organisms of Section 3.3 do the work, given weeks instead of hours and sunlight instead of blowers.

3.13.2 The pond series

Pond	Depth (m)	Retention (d)	Function
Anaerobic	2 – 5	1 – 5	Bulk BOD removal by digestion; 50–70%. The cheapest removal available
Facultative	1 – 2	5 – 30	Main BOD removal by the algal–bacterial partnership
Maturation	1 – 1.5	5 – 20	Pathogen destruction and polishing

Table 3.15 The conventional pond series. Total retention 20–60 days.

3.13.3 The facultative pond and the algal–bacterial partnership

The facultative pond is the heart of a pond system, and the partnership within it is worth understanding properly.

Sunlight penetrates the upper layer, allowing ALGAE to grow. In photosynthesis algae take up carbon dioxide and release oxygen. The BACTERIA in the same layer do the reverse: they consume oxygen while oxidising the organic matter, and release carbon dioxide. Each supplies precisely what the other requires.



Meanwhile solids settle to the floor where, in the absence of oxygen, they digest anaerobically. A facultative pond is thus three reactors stacked vertically: aerobic at the top, facultative in the middle, anaerobic at the bottom.

Because the oxygen supply depends on photosynthesis, and photosynthesis on sunlight, these ponds are designed on SURFACE AREA rather than volume:

$$\lambda_s = 10 L_0 Q / A \quad (\text{kg BOD/ha}\cdot\text{d})$$

L_0 = influent BOD (mg/L); Q = flow (m^3/d); A = pond area (m^2)

The permissible loading rises with temperature; a widely used relation is that of Mara:

$$\lambda_s = 350 (1.107 - 0.002 T)^{(T - 25)}$$

T = mean air temperature of the coldest month ($^{\circ}\text{C}$)

KEY IDEA

In a facultative pond, sunlight supplies the oxygen through the algae. This is why ponds are designed on AREA and why their performance depends on climate — and why deepening a facultative pond adds volume but no extra capacity.

WORKED EXAMPLE 3.6 — Pond system design

Design an anaerobic pond followed by a facultative pond for $300 \text{ m}^3/\text{d}$ of wastewater at BOD 500 mg/L . The coldest-month temperature is 20°C .

PART A — ANAEROBIC POND

Step 1 BOD load = $0.500 \times 300 = 150 \text{ kg/d}$

Step 2 At a volumetric loading of $250 \text{ g/m}^3\cdot\text{d} = 0.250 \text{ kg/m}^3\cdot\text{d}$: $V = 150/0.250 = 600 \text{ m}^3$

Step 3 HRT = $600/300 = 2.0 \text{ d}$ (within the usual 1–5 d)

Step 4 At 3 m depth, area = 200 m^2 — say $20 \text{ m} \times 10 \text{ m}$

Step 5 At 60% removal, effluent BOD = 200 mg/L ; load carried forward = 60 kg/d

PART B — FACULTATIVE POND

Step 6 Permissible areal loading at 20°C :

$$\lambda_s = 350(1.107 - 0.040)^{(20-25)} = 350(1.067)^{(-5)}$$

$$(1.067)^5 = 1.382, \text{ so } (1.067)^{-5} = 0.7236$$

$$\lambda_s = 350 \times 0.7236 = 253 \text{ kg BOD/ha}\cdot\text{d}$$

Step 7 Area = $10 \times 200 \times 300 / 253 = 600,000/253 = 2,372 \text{ m}^2$, say $2,400 \text{ m}^2$ (0.24 ha)

Step 8 At 1.5 m depth, $V = 3,600 \text{ m}^3$; HRT = 12 days (within 5–30 d)

SUMMARY: anaerobic pond 600 m^3 (3 m deep, $20 \times 10 \text{ m}$, 2 d); facultative pond $2,400 \text{ m}^3$ (1.5 m deep, 12 d). Total land about 0.28 ha.

COMMENT: compare with Worked Example 3.1, where a comparable load was treated in a tank of 420 m^3 . The pond system needs roughly ten times the land — but no blowers, no clarifier, no return sludge pumping and virtually no electricity. That is the trade this section describes.

3.13.4 Aerated lagoons

A basin 3–5 m deep fitted with mechanical aerators. Oxygen no longer depends on algae, so the loading can be far higher and the land requirement falls sharply. Two kinds are distinguished by the power supplied: a FULLY-MIXED lagoon keeps all solids in suspension and behaves like activated sludge without sludge return (retention 3–10 days); a FACULTATIVE lagoon supplies oxygen but lets solids settle and digest on the floor.

An important kinetic consequence follows from the absence of recycle: the sludge age equals the hydraulic retention time. Because SRT can no longer be set independently, the retention must be long enough by itself to prevent washout — hence 3–10 days rather than the 4–8 hours of activated sludge. A settling pond must follow, since biomass leaves with the effluent.

3.13.5 Floating aquatic plant systems

A shallow pond covered with floating macrophytes — most commonly water hyacinth (*Eichhornia crassipes*) or duckweed (*Lemna* species). Both grow abundantly in Bangladesh.

It is a common misconception that the plants do the treating. They contribute, but most of the work is done by bacteria; the plants' principal role is to create the conditions in which those bacteria thrive.

- **The root mass is a biofilm support.** A water hyacinth develops a dense fibrous root system hanging half a metre into the water, providing an enormous attached-growth surface. Most BOD removal occurs there — the system is in effect an attached-growth reactor whose media happen to be alive.
- **Oxygen transport.** The plants carry oxygen from the atmosphere through their tissues to the root zone.

- **Nutrient uptake.** The plants take up nitrogen and phosphorus for their own growth — significant quantities, PROVIDED the plants are harvested.
- **Shading.** The cover suppresses algal growth, so the effluent does not carry the algal solids that spoil a maturation pond's suspended-solids figure.
- **Metal uptake.** Water hyacinth accumulates heavy metals including chromium — of obvious interest for tannery effluent, and equally obviously meaning the harvested plants become contaminated material.

Parameter	Water hyacinth	Duckweed
Depth	0.5 – 1.5 m	0.4 – 1.5 m
Retention	5 – 15 d	10 – 25 d
BOD loading	30 – 80 kg/ha·d	20 – 60 kg/ha·d
BOD removal	70 – 90%	60 – 85%
Nitrogen removal	50 – 80% with harvesting	40 – 75%
Harvesting	Weekly to fortnightly	Every few days

HARVESTING IS PART OF THE PROCESS

Water hyacinth doubles its biomass in roughly a fortnight. If it is not harvested, the mat thickens, the lower plants die and sink, and their decomposition consumes oxygen and releases the very nutrients that had been taken up. An unharvested system becomes a source of pollution rather than a sink.

Harvesting is therefore not maintenance but part of the process, and the labour it requires is the principal operating cost.

3.13.6 Assessment of natural systems

Advantages	Limitations
Lowest capital and operating cost of any process	Very large land requirement — usually decisive
Little or no energy	Performance varies with season and sunlight
Very simple; minimal skilled supervision	Long retention: 20–60 days for a full series
Excellent pathogen removal in maturation ponds	Algae leave with the effluent, raising SS and BOD

Advantages	Limitations
No daily sludge handling (desludge every 2–10 y)	Odour from anaerobic ponds, especially with sulphate
Absorbs shock loads by sheer volume	Little control once built; poor for toxic industrial loads

3.14 Anaerobic Treatment

3.14.1 Why treat anaerobically

For a strong wastewater, removing oxygen from the equation changes the economics entirely.

- **No aeration energy.** Aeration is 45–60% of an aerobic plant's energy bill; anaerobic treatment eliminates it.
- **It PRODUCES energy.** The organic load is converted largely into methane, which can be burned for heat or power.
- **Very little sludge.** Yield is 0.05–0.15 kg VSS per kg COD against 0.4–0.6 aerobically — sludge handling costs collapse.
- **It handles high strength.** Aerobic systems struggle above about 1,500–2,000 mg/L COD because the oxygen demand becomes impractical; anaerobic systems thrive there.

The catch is that anaerobic treatment does NOT produce a dischargeable effluent. It is a pre-treatment: an aerobic polishing stage almost always follows.

3.14.2 The four-stage chain

Anaerobic conversion is not one reaction but a chain of four, carried out by different groups of organisms in sequence.

Stage	Conversion	Organisms
1. Hydrolysis	Polymers → soluble monomers	Extracellular enzymes of fermentative bacteria
2. Acidogenesis	Monomers → volatile fatty acids, alcohols, H ₂ , CO ₂	Fast-growing fermentative bacteria
3. Acetogenesis	Longer VFAs → acetate, H ₂ , CO ₂	Hydrogen-producing acetogens
4. Methanogenesis	Acetate and H ₂ /CO ₂ → CH ₄ + CO ₂	Methanogenic archaea — SLOW and sensitive

Table 3.16 The four stages of anaerobic digestion. About 70% of the methane comes from acetate and 30% from hydrogen.

3.14.3 The stability problem

The kinetic difference between the groups is the source of every operating difficulty. Acidogens have maximum growth rates of the order of 2 d^{-1} ; methanogens perhaps $0.2\text{--}0.4 \text{ d}^{-1}$ — an order of magnitude slower.

If the organic loading rises suddenly, the acidogens respond at once and produce volatile fatty acids rapidly. The methanogens cannot accelerate correspondingly. Acids accumulate, consume the bicarbonate alkalinity, and the pH falls. Methanogens are inhibited below about pH 6.5, so they slow further and more acid accumulates. The process is self-accelerating, and it is called SOURING.

The quantitative indicator is the ratio of volatile acids to alkalinity:

$$\text{VFA} / \text{Alkalinity} < 0.3 \quad (\text{stable})$$

both as mg/L, alkalinity as CaCO_3 ; above 0.5 the reactor is in difficulty

KEY IDEA

In an anaerobic reactor the pH is a LAGGING indicator. By the time it falls, the alkalinity is spent and recovery is difficult. The VFA/alkalinity ratio moves first, and is what an operator should watch daily.

3.14.4 Operating conditions

Parameter	Optimum	Why it matters
pH	6.8 – 7.4	Methanogens inhibited below 6.5
Temperature (mesophilic)	30 – 38 °C	The usual band; Bangladesh's climate suits it
Alkalinity	2,000 – 4,000 mg/L as CaCO_3	Buffers the acids produced
VFA / alkalinity	< 0.3	The early-warning indicator
Nutrients COD : N : P	≈ 350 : 5 : 1	Far less N needed than aerobically
Sulphide	< 200 mg/L	Directly relevant to tannery effluent

3.14.5 Reactor types

Reactor	Principle	Loading (kg COD/m ³ ·d)
Anaerobic pond	Open basin, long retention	0.1 – 0.4
Conventional digester	Stirred tank; biomass with the liquid	1 – 3
Anaerobic filter	Biofilm on fixed media	5 – 15
Anaerobic baffled reactor (ABR)	Baffles force flow up and down through settled sludge blankets	3 – 10
UASB	Granular sludge blanket, upflow	5 – 15 (up to 25)

The anaerobic baffled reactor

A rectangular tank divided by vertical baffles into four to eight compartments, arranged alternately as down-comers and up-comers so that the wastewater is forced DOWN under one baffle and UP through the next, passing through a settled sludge blanket in each compartment.

Three features give it its character. Biomass is retained with no moving part and no gas–liquid–solid separator. Construction is simple and operation needs little skill. And — the important kinetic point — acidogenesis dominates in the first compartments where substrate is abundant, while methanogenesis dominates in the later ones. The two groups, difficult to reconcile in one vessel, are thereby SEPARATED IN SPACE, each operating near its own optimum.

The UASB reactor

The upflow anaerobic sludge blanket reactor achieves higher loadings by cultivating the biomass as dense GRANULES. Under a steady upflow the biomass self-assembles into granules 1–3 mm across, containing all four trophic groups and settling at 30–80 m/h. Because they settle so fast they are retained even at high upflow velocities, so an enormous biomass is held in a small volume.

At the top, a GLS (gas–liquid–solid) separator collects biogas in an inverted hood, allows granules to fall back, and lets clarified effluent leave. Without it, rising gas bubbles would carry the granules out. Typical performance is 70–85% COD removal at 5–15 kg COD/m³·d with an HRT of only 6–12 hours. Granulation takes two to six months unless the reactor is seeded.

3.14.6 Biogas

From stoichiometry, the theoretical methane yield is:

$$0.35 \text{ m}^3 \text{ CH}_4 \text{ per kg of COD stabilised} \quad (\text{at } 0^\circ \text{C, } 1 \text{ atm})$$

rising to about 0.40 m³/kg at 35 °C. Biogas is 60–70% methane and 30–40% carbon dioxide, with a calorific value of 21–25 MJ/m³ — about 60% that of natural gas. It can fire a boiler, run an engine or be upgraded.

THE H₂S PROBLEM — CRITICAL FOR TANNERIES

Tannery effluent is rich in sulphate (from pickling and chrome tanning) and sulphide (from liming). In the reactor, sulphate-reducing bacteria convert sulphate to sulphide, which partitions into the gas as H₂S.

This does two kinds of damage. H₂S burns to sulphur dioxide and forms sulphuric acid, corroding engines, boilers, gas lines and concrete. And sulphide is directly TOXIC to methanogens above roughly 200 mg/L. Sulphate-reducers also compete for the same substrate, so a high-sulphate effluent yields LESS methane per kg COD.

Tannery biogas must therefore be scrubbed before use — and this is another reason why sulphide oxidation in Chapter 1 matters.

3.14.7 Anaerobic versus aerobic

Feature	Anaerobic	Aerobic
Energy	Produces (methane)	Consumes (aeration)
Sludge yield	0.05 – 0.15 kg VSS/kg COD	0.4 – 0.6 kg VSS/kg COD
Best influent strength	High (> 1,500–2,000 mg/L COD)	Low to medium
Effluent quality	Partial — needs polishing	Meets discharge standards
Start-up time	2 – 6 months (granulation)	2 – 4 weeks
Temperature sensitivity	High — needs 30–38 °C	Moderate
Robustness to toxins	Lower — methanogens sensitive	Higher

3.15 Biological Nitrogen Removal

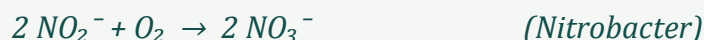
3.15.1 Why nitrogen must be removed

Ammonia in an effluent does three kinds of harm. It is directly toxic to fish in its un-ionised form at concentrations below 1 mg/L; it exerts a nitrogenous oxygen demand of 4.57 kg O₂ per kg N in the receiving water; and with phosphorus it causes eutrophication, the algal blooms whose decay strips the oxygen from a watercourse.

For a tannery the matter is pressing, because deliming with ammonium salts and the breakdown of hide protein make tannery effluent unusually rich in ammonia nitrogen.

3.15.2 Nitrification

Nitrification proceeds in two steps carried out by two genera of AUTOTROPHIC bacteria — organisms that take their carbon from carbon dioxide and their energy from the nitrogen itself.



Overall: $\text{NH}_4^+ + 2 \text{O}_2 \rightarrow \text{NO}_3^- + 2 \text{H}^+ + \text{H}_2\text{O}$. Three consequences of design importance follow from the stoichiometry.

Quantity	Value	Derivation
Oxygen required	4.57 kg O ₂ /kg N	2 mol O ₂ (64 g) per mol N (14 g)
Alkalinity destroyed	7.14 kg CaCO ₃ /kg N	2 mol H ⁺ released per mol N
Cell yield	0.10 – 0.15 kg VSS/kg N	Low — autotrophic growth

The second row is frequently overlooked and causes real failures. Destroying 7.14 kg of alkalinity per kg of nitrogen will exhaust the buffering capacity of a poorly-buffered wastewater; the pH falls; and since nitrifiers are inhibited below pH 6.8, nitrification stops. The plant then appears to fail for no visible reason.

WORKED EXAMPLE 3.7 — Alkalinity check

A plant treats 500 m³/d containing 60 mg/L of ammonia nitrogen. Influent alkalinity is 350 mg/L as CaCO₃. Will nitrification be sustained?

Step 1 Nitrogen oxidised = 0.060 × 500 = 30 kg N/d

Step 2 Alkalinity destroyed = $7.14 \times 30 = 214 \text{ kg CaCO}_3/\text{d}$

Step 3 As a concentration: $214/500 = 0.428 \text{ kg/m}^3 = 428 \text{ mg/L}$

Step 4 Alkalinity available = 350 mg/L

CONCLUSION: demand (428) exceeds supply (350). The alkalinity will be exhausted, the pH will collapse and nitrification will cease.

REMEDY: dose alkali — the shortfall including a residual of 100 mg/L is about 178 mg/L , that is 89 kg/d as CaCO_3 — or incorporate denitrification, which returns about half the alkalinity consumed. The second is preferable: it costs nothing in chemicals and removes the nitrogen as well.

3.15.3 The sludge age requirement

Nitrifiers grow slowly, and since $1/\text{SRT}$ equals the net growth rate, the minimum sludge age is fixed by that growth rate. Temperature correction is essential.

Temperature	$\mu_m \text{ (d}^{-1}\text{)}$	Minimum SRT (d)	Design SRT (SF 2.5)
15 °C	0.30	4.0	10
20 °C	0.45	2.5	6.3
25 °C	0.65	1.7	4.3
30 °C	0.90	1.2	3.0

Table 3.17 Sludge age required for nitrification (decay coefficient 0.05 d^{-1}).

The table shows the climate advantage plainly. A plant in Bangladesh at $25\text{--}30^\circ\text{C}$ needs a sludge age of three to four days to nitrify, where the same plant in a temperate winter would need seventeen. This is why the rule of thumb quoted earlier — SRT of about ten days — is conservative for local conditions.

3.15.4 Denitrification

Denitrification is carried out by ORDINARY HETEROTROPHIC bacteria which, deprived of oxygen but supplied with nitrate, use the nitrate as their electron acceptor:



Three consequences follow, and all three are favourable.

- **Nitrogen leaves as a GAS.** It is not transferred to the sludge; it escapes to an atmosphere that is already 78% nitrogen.
- **Alkalinity is recovered** — about 3.57 kg CaCO₃ per kg N, roughly half that destroyed by nitrification. This is what makes the combination self-buffering.
- **Oxygen is recovered** — using nitrate as the acceptor recovers about 2.86 kg of oxygen equivalent per kg N, a genuine reduction in aeration.

The carbon requirement is about 4 kg BOD per kg of nitrate nitrogen reduced. If the incoming wastewater's own BOD is used this costs nothing; if it has already been oxidised, methanol must be purchased. This economic fact dictates the process arrangement.

3.15.5 Process configurations

Arrangement	Description
Post-denitrification	Anoxic zone AFTER the aerobic zone. Denitrification is complete, but the carbon has already been consumed, so an external source must be bought. Rarely economical
Pre-denitrification (MLE)	Anoxic zone BEFORE the aerobic zone, with nitrified mixed liquor recycled back to it. The influent's own BOD serves as the carbon source, at no cost. THE STANDARD ARRANGEMENT
Four-stage Bardenpho	Anoxic–aerobic–anoxic–aerobic. The second anoxic zone removes nitrate that escaped the first, using endogenous carbon. Total nitrogen below 3 mg/L achievable
Simultaneous nitrification–denitrification	In an oxidation ditch, DO is high just after each rotor and near zero further round the loop, so both occur in the same channel with no recycle pumping. DO must be held at about 0.3–0.7 mg/L

For the MLE arrangement, nitrogen removal is limited by how much nitrate can be recycled:

$$\text{Removal efficiency} = (R_i + R_s) / (1 + R_i + R_s)$$

R_i = internal recycle ratio; R_s = sludge return ratio

Since some nitrate always leaves with the effluent, complete removal is impossible in a single-stage MLE. In practice such plants are seldom designed beyond about 80% removal, because the pumping cost rises very steeply thereafter.

KEY IDEA

Nitrification and denitrification are complementary in three separate ways. Denitrification returns half the alkalinity nitrification destroyed; it recovers part of the oxygen nitrification consumed; and it uses the incoming BOD that would otherwise have to be oxidised with purchased air.

Designing them together is not merely tidier — it is markedly cheaper than designing either alone.

3.16 Biotechnology in Effluent Treatment

3.16.1 What is meant here

Every process in this chapter is already biological. What distinguishes BIOTECHNOLOGY in the present sense is that we no longer merely provide conditions and let whatever organisms arrive do the work. Instead we intervene deliberately — selecting particular organisms, adding cultivated strains, supplying enzymes, or engineering the physical form the biomass takes.

Conventional biological treatment asks: what conditions will encourage a useful community to establish itself? Biotechnology asks: which organisms do I want, and how do I put them there and keep them?

3.16.2 Bioaugmentation

Bioaugmentation. the deliberate addition of selected or cultivated micro-organisms to a treatment system, supplied as dried or liquid cultures and dosed into the reactor.

It is applied in four situations: accelerating the start-up of a new plant; restoring activity after a toxic shock; degrading specific recalcitrant compounds such as particular syntans or surfactants; and controlling nuisances such as filamentous bulking or grease accumulation.

A realistic assessment is necessary. Bioaugmentation works best where the wastewater contains a specific compound the resident population cannot degrade, and where the added organisms find conditions they can survive in. It is much less effective as a general tonic: if the underlying problem is low dissolved oxygen, nutrient deficiency or a toxic influent, no dosed culture will compensate. Correct the conditions first.

3.16.3 Acclimatised and halotolerant biomass

For tannery effluent this is the most important biotechnological approach, because it addresses the industry's characteristic difficulty.

High dissolved-salt concentrations impose osmotic stress on micro-organisms, drawing water out of the cells. Above roughly 10,000–15,000 mg/L of TDS a conventional activated sludge deteriorates badly, the flocs dispersing and the effluent turning turbid.

Two facts about salinity are frequently misunderstood and worth emphasising.

1. **Steady high salinity is far less damaging than FLUCTUATING salinity.** A community given time will adapt; the same community subjected to sudden swings will not. Equalisation therefore protects the biology as effectively as any dosed culture.
2. **Adaptation can be cultivated deliberately.** By raising the salt concentration gradually over several weeks, a HALOTOLERANT community is selected which continues to function at concentrations that would destroy an unadapted sludge.

Attached-growth systems have an inherent advantage in saline effluent, since the biomass is physically fixed and cannot be washed out even when stressed.

3.16.4 Enzymes and immobilised biomass

ENZYMES are the catalysts micro-organisms use to break down complex molecules; they are proteins, not living organisms. Adding them directly can accelerate the rate-limiting hydrolysis step. In leather-sector effluent the relevant preparations are LIPASES, which break down the fats and greases from fleshing and fatliquoring, and PROTEASES, which hydrolyse the protein load from hair and hide. Since hydrolysis is often the slowest step, assisting it can genuinely improve performance — but enzymes are consumed and denatured, so dosing must be continuous.

IMMOBILISED CELLS are micro-organisms entrapped in a gel or fixed to a carrier so that a chosen population is retained regardless of hydraulic conditions. The carrier systems of Section 3.12 are the industrial realisation of this idea. AEROBIC GRANULAR SLUDGE takes it further: biomass cultivated in batch cycles with a very short settling period self-assembles into dense granules, each aerobic outside and anoxic within, removing carbon and nitrogen simultaneously and settling in minutes.

3.16.5 Biotechnology for resource recovery

The most recent direction reframes the treatment plant as a place where materials are recovered rather than merely destroyed. Anaerobic processes produce methane, which is a fuel. Biological phosphorus removal concentrates phosphate in the sludge, from which struvite fertiliser may be recovered. And in the leather sector specifically, enzymatic hydrolysis of chrome shavings and trimmings yields collagen hydrolysate and gelatin, converting a hazardous solid waste into a saleable product.

3.16.6 A balanced view

- **Conditions dominate.** No organism performs if the dissolved oxygen is inadequate, the nutrients deficient or the influent toxic.
- **Added organisms must survive.** A dosed culture competes with an established resident population; unless it is genuinely better suited to the conditions it will simply be washed out.
- **Prevention outperforms treatment.** Reducing salt at curing, exhausting chrome more completely, or unhairing with enzymes instead of sulphide reduces every downstream load at a fraction of the cost.

3.17 Comparison of Biological Processes

Process	Energy	Footprint	Skill	Best suited to
Activated sludge	High	Low	High	Reliable quality; compact site
Extended aeration	High	Moderate	Moderate	Industrial effluent; least sludge
Oxidation ditch	Moderate	Moderate	Low–mod	Robust; simultaneous N removal
SBR	High	Low	High	Variable flow; nutrient removal
Trickling filter	Very low	Large	Low	Low-cost; shock-tolerant
RBC	Low	Compact	Low	Small–medium flows; staged nitrification
MBBR / IFAS	Moderate	Compact	Moderate	Upgrading; toxic and variable loads
MBR	High	Very low	High	Reuse; strict limits; no land
Ponds / lagoons	Very low	Very large	Very low	Low budget; warm climate; land available
Aquatic plant systems	Very low	Large	Low	Polishing; nutrient removal
UASB / ABR	Produces	Low	Moderate	High-strength effluent; biogas

Table 3.18 Comparison of biological processes.

Reading this table, three groupings emerge. The MECHANISED AEROBIC processes buy compactness with energy and operator skill. The NATURAL systems buy cheapness with land and time. The ANAEROBIC processes buy energy recovery at the price of an effluent that still needs polishing. There is no universally best choice — only a choice appropriate to a particular set of constraints.

3.18 Tannery-Specific Considerations and Process Selection

3.18.1 The four constraints

Constraint	Why it hurts the biology	Where it is solved
High salinity (TDS)	Osmotic stress disperses flocs; biological treatment cannot remove salt at all	Acclimatised biomass (3.16); salt reduction at source; RO in Chapter 2
Sulphide	Toxic to aerobes and to methanogens; exerts oxygen demand; corrodes	Oxidation in PRIMARY treatment (Chapter 1)
Chromium	Heavy-metal inhibition; contaminates sludge as hazardous waste	Segregation and recovery in PRIMARY treatment (Chapter 1)
High and variable load	Shock loads overwhelm or poison the biomass	Equalisation (Chapter 1); robust, forgiving process choice

THE CENTRAL LESSON

The success of biological treatment is largely decided UPSTREAM. Sulphide oxidation, chrome recovery, stream segregation and equalisation are primary-treatment duties — but they are what keep the biological stage alive.

A biological design may be faultless and still fail if the primary stage does not deliver a clean, steady, non-toxic influent.

3.18.2 Choosing the process

A practical decision sequence, rather than a formula:

1. **How strong is the effluent?** Above roughly 2,000 mg/L COD, put an anaerobic stage (UASB, ABR or anaerobic pond) first to strip the load and recover biogas.
2. **How much land is available?** Abundant land and a low budget favour ponds and lagoons; constrained land favours extended aeration, MBBR or MBR.
3. **How skilled is the operating team?** Limited skilled staff strongly favours attached growth or extended aeration over closely-controlled activated sludge.

4. **How variable and toxic is the influent?** Frequent shocks favour biofilm systems, which recover far faster than suspended sludge.
5. **Is ammonia limited? Is reuse required?** Ammonia limits demand a long sludge age and an anoxic zone; water reuse demands an MBR or UF followed by RO.

3.18.3 The usual answer

For a typical tannery the train that has emerged from practice is:

*Equalisation → (anaerobic stage) → extended-aeration or MBBR aerobic stage
→ secondary clarifier → tertiary polishing*

Extended aeration is chosen so often for good reasons: the influent is strong, so a large tank is needed anyway; the influent is variable and sometimes hostile, and a large biomass with long retention absorbs shocks; sludge handling is expensive, and endogenous respiration reduces sludge production to a third or less; ammonia must be removed, and the long sludge age nitrifies without further provision; and operators may be few and unskilled, and the process is forgiving. The price is a large tank and higher aeration energy — usually worth paying.

A NOTE ON COMMON EFFLUENT TREATMENT PLANTS

Where many small tanneries share a CETP, the biological stage is designed on the same principles but faces an additional difficulty: the influent is the sum of many uncoordinated discharges, and pre-treatment at source is often not enforced. Chromium and sulphide then arrive already mixed, and no biological design can compensate.

A CETP's biological stage can only be as good as the pre-treatment of the factories feeding it.

3.19 Chapter Summary

The main points

1. Biological treatment removes the dissolved biodegradable organic matter that neither physical nor chemical processes can remove economically. It is microbial farming: grow the organisms, feed them the pollution, harvest them as sludge.
2. Every biological process is a combination of two choices — aerobic or anaerobic, suspended or attached growth. There are only four possibilities.
3. The control parameters are MLSS/MLVSS, HRT, SRT, F/M and DO, with SVI as the diagnostic. The sludge wasting rate controls MLSS and SRT together, and is the operator's most powerful lever.
4. $SRT \gg HRT$: sludge is retained for days while water passes in hours. Sludge age also SELECTS THE SPECIES, which is why nitrification requires a long SRT.
5. Activated sludge is the reference process; its modifications are simply different settings of the F/M – SRT dial, or different tank shapes.
6. Aeration serves two duties — oxygen supply and mixing — and is 45–60% of plant energy. Hold DO at about 2 mg/L.
7. Attached-growth systems fix the biomass so it cannot wash out: no sludge return, no bulking, better tolerance of shock, but less control.
8. Ponds and aquatic plant systems trade land and time for energy and machinery; harvesting is part of the process, not maintenance.
9. Anaerobic treatment produces energy instead of consuming it and yields little sludge, but gives only partial treatment and is delicate. Watch the VFA/alkalinity ratio, not the pH.
10. Nitrification and denitrification are complementary: the second returns half the alkalinity and part of the oxygen the first consumed, and uses the influent's own BOD.
11. For tannery effluent the constraints are salinity, sulphide, chromium and variability — and three of the four are solved upstream in primary treatment.

Where the three chapters have brought us

Parameter	Raw	After Ch. 1	After Ch. 3
BOD ₅ (mg/L)	1,000 – 3,000	800 – 1,200	50 – 150
COD (mg/L)	3,000 – 8,000	2,500 – 4,000	200 – 500
Suspended solids	2,000 – 4,000	400 – 800	50 – 150
Sulphide	100 – 300	< 2	< 2
Chromium	100 – 400	Recovered	—
Ammonia N	High	High	Low if nitrifying
Dissolved solids	10,000 – 40,000	Unchanged	Unchanged

Table 3.19 The cumulative effect of the three chapters (Chapter 2 processes applied selectively as required).

The last row is the honest conclusion of this whole study material. Three chapters of technology have reduced the organic load by more than ninety-five per cent, destroyed the sulphide, recovered the chromium and clarified the water — and have left the dissolved salts exactly as they were found. Only reverse osmosis (Chapter 2) removes them, at high cost, and only reduced salt use in the tannery itself avoids the problem altogether.

THE CENTRAL LESSON OF THE WHOLE MATERIAL

Treatment does not destroy pollution. It separates it, transforms it, or converts it into biomass — and what is removed from the water becomes sludge, spent carbon, brine or salt, all of which must still be managed.

The cheapest and cleanest litre of effluent is always the one that was never polluted in the first place. Cleaner production — low-salt curing, high-exhaustion chrome tanning, hair-save unhairing, reduced water use — lowers every load described in these three chapters at a fraction of the cost of removing it afterwards.

Review questions

1. Why can the dissolved biodegradable load not be removed economically by adsorption, oxidation or membranes?
2. Write the overall aerobic reaction and identify which product must be physically removed, and why.

3. State the nutrient ratio required for aerobic growth, and explain what happens if phosphorus is deficient.
4. State the two questions that classify every biological process, and place the oxidation ditch, trickling filter, UASB and RBC in the grid.
5. Distinguish carefully between anoxic and anaerobic conditions.
6. Distinguish MLSS from MLVSS, stating how each is measured and which is used in the F/M ratio.
7. Explain why SRT is much greater than HRT, and why sludge age determines which organisms survive.
8. A plant treats $800 \text{ m}^3/\text{d}$ of BOD 450 mg/L at F/M 0.25 with MLVSS $2,000 \text{ mg/L}$. Calculate the aeration tank volume and the HRT.
9. An SV_{30} test gives 340 mL/L with MLSS $2,600 \text{ mg/L}$. Calculate the SVI and comment.
10. Explain the three stages of removal within an aeration tank, and why contact time matters.
11. Why is extended aeration so often chosen for tannery effluent? Give four reasons.
12. Calculate the oxygen requirement for a plant removing 320 kg BOD/d and oxidising $25 \text{ kg NH}_4\text{-N/d}$, and estimate the blower power at $1.5 \text{ kg O}_2/\text{kWh}$.
13. Why is a dissolved oxygen concentration of 4 mg/L wasteful in two separate ways?
14. List, in the order you would apply them, the remedies for filamentous bulking.
15. Why does the dissolved oxygen RISE during a toxic shock, and what should the operator do first?
16. What is sloughing, and to what does it correspond in an activated sludge plant?
17. Design a rock trickling filter for $500 \text{ m}^3/\text{d}$ at BOD 300 mg/L to achieve 80% removal, with $R = 2.0$ and depth 2.0 m . Classify the resulting filter.
18. An RBC treats $300 \text{ m}^3/\text{d}$ at BOD 250 mg/L with nitrification at $3 \text{ g BOD/m}^2\cdot\text{d}$. Determine the disc area and check the first-stage loading.
19. A 500 m^3 tank at MLVSS $2,800 \text{ mg/L}$ must accept an increase from 300 to 450 kg BOD/d at constant F/M. Using carriers giving 6 kg VSS/m^3 , find the carrier volume and fill fraction.
20. Describe the algal–bacterial partnership in a facultative pond, and explain why ponds are designed on area.
21. Explain why harvesting is part of the process in an aquatic plant system.

22. Name the four stages of anaerobic digestion and explain, with reference to growth rates, why souring occurs.
23. Why must tannery biogas be scrubbed before use?
24. A plant oxidises 45 kg N/d at 600 m³/d with influent alkalinity 300 mg/L. Check whether nitrification is sustainable.
25. Explain why pre-denitrification is preferred to post-denitrification.
26. A tannery has a strong, saline, variable effluent and unskilled operators. Recommend a biological process train with reasons.

References

- Metcalf & Eddy — Wastewater Engineering: Treatment and Resource Recovery. Principal source of design parameters and process theory throughout.
- Grady, C. P. L., Daigger, G. T., Love, N. G. and Filipe, C. D. M. — Biological Wastewater Treatment.
- Rittmann, B. E. and McCarty, P. L. — Environmental Biotechnology: Principles and Applications. Biofilm theory and nutrient removal.
- Jenkins, D., Richard, M. G. and Daigger, G. T. — Manual on the Causes and Control of Activated Sludge Bulking and Foaming.
- Arceivala, S. J. and Asolekar, S. R. — Wastewater Treatment for Pollution Control and Reuse.
- Mara, D. — Domestic Wastewater Treatment in Developing Countries. Pond design and the areal loading relation.
- van Haandel, A. and Lettinga, G. — Anaerobic Sewage Treatment: A Practical Guide for Warm Climates.
- Scholz, W. — Key Technologies for Tannery Wastewater Treatment (IULTCS); UNIDO — Introduction to Treatment of Tannery Effluents.
- Bangladesh Department of Environment — Environmental Conservation Rules (ECR) 2023.